TRAFFIC SIGN CLASSIFICATION WITH GOOGLENET AND COLOR FILTER LAYER

Siyu Shao
Student# 1007147204
jasmine.shao@mail.utoronto.ca

Xinghan Yan
Student# 1005678901
xinghan.yan@mail.utoronto.ca

Pengzhou Xu
Student# 1006732074
pengzhou.xu@mail.utoronto.ca

Weite Fan
Student# 1006761933
alan.fan@@mail.utoronto.ca

—-Total Pages: 9

1 INTRODUCTION

Traffic sign recognition (TSR) has become a key technology as autonomous driving and advanced driver assistance systems (ADAS) continue to evolve. Traffic signs provide essential vital information for both human drivers and autonomous systems to drive safely and efficiently, such as speed limits, warnings, and road conditionsKim et al. (2023). Accurate recognition of traffic signs in real time not only provides assistance for drivers but also improves the overall safety and reliability of modern transportation systems by implementing traffic rulesTang (2024).

Our project aims to develop a deep learning-based system for real-time traffic sign recognition with high accuracy. Traffic signs are typically distinguished by their shape and color, so our approach will utilize a convolutional neural network (CNN) architecture that is designed to capture these features effectively. Using CNN, we will detect and classify traffic signs from real-world images of the GTSRB dataset, with reliable performance even in challenging data qualities such as poor lighting, occlusions, and image distortions.

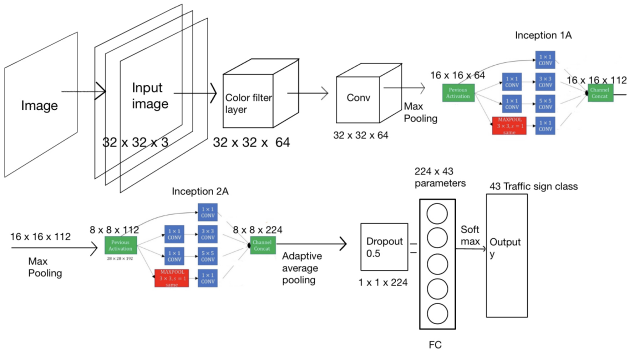


Figure 1: Illustration of the overall pipeline of our model.

The need for smarter, safer driving solutions are addressed using the work contained in this project. During interactions with traffic signs, misinterpretation by drivers or automated systems can adversely affect driver behavior, leading to collisions or violations. Deep learning can automate TSR, reduce human error, improve real-time decision making, and make for safer road transportation.

Traffic Sign Recognition (TSR) can be successfully carried out by deep learning which means that it can automatically learn and extract the right type of features based on the image. The results are

robust and accurate under challenging conditions like change in weather, occlusion or lighting Li et al. (2023).

We will use CNNs which have been optimized to recognize images, for our TSR project. CNNs are particularly effective at processing grid structured data and are computationally and accurate for real time applications like TSR making them ideally suited for real world problems in autonomous driving and anywhere there is a need for rapid decision makingTang (2024).

2 Background and Related Work

Traffic Sign Recognition (TSR) is a critical technology that improves the transportation system. The development of TSR systems dates back to the early 2000s. Numerous methods have been implemented since the past few decades to deal with this problem. Initially, people relied on color and shape recognition for traffic sign detection. With the emergence of deep learning methods, these techniques have also been applied to real-time traffic sign detection.

2.1 Classical color and shape recognition

This method typically relies on extracting low-level features like color, shape, and edges to detect traffic signs. One of the earlier significant contributions to Traffic Sign Recognition (TSR) came from X.W. Gao, who proposed a method based on human vision models for the extraction of color and shape features of traffic signs. Their approach utilized color information to segment traffic signs from the background, followed by shape detection techniques to classify them into different categories. The recognition rate is up to 95% Gao et al. (2006). However, this method is easily affected by weather conditions in the input image, and that will influence image brightness.

2.1.1 Deep learning using Neural Networks

With the advent of deep learning and the availability of larger datasets, Convolutional Neural Networks (CNN) took the lead in this research field. For example, Xiaoxuan Qiao introduced the CNN deep learning network to conduct training while identifying features, effectively improving recognition accuracy Qiao (2023). The combination of fixed and learnable layers and the technique of bootstrap can improve the model accuracy and avoid overfitting problems Wu et al. (2013).

2.2 Real-time traffic sign recognition

Faster Region-based Convolutional Neural Network (R-CNN) and You Only Look Once (YOLO) models are popular for real time data analysis. Njayou Youssouf effectively demonstrates the application of both Faster R-CNN and YOLOv4 for traffic sign detection Youssouf (2022). Since YOLO suffers from false detection and missed detection on small objects, advanced YOLO models like YOLOv5-TS are applied to extract multi-scale features at the same time prevent feature loss Shen et al. (2023)

3 Data Processing

The German Traffic Sign Recognition Benchmark (GTSRB) is a comprehensive dataset curated for traffic sign recognition tasks, making it a cornerstone in the fields of computer vision and autonomous driving. Introduced in 2011 as part of a multi-class classification competition, the dataset comprises over 50,000 images of traffic signs captured under real-world conditions.

The raw dataset was initially downloaded in a zipped format from Kaggle. A Python script is developed to locally randomize and balance the dataset. This approach significantly reduced processing time and allowed for more efficient data management. In total 150 training samples and 30 validation samples are taken per class across all 43 classes, ensuring a 70-15-15 data split. This balanced dataset enhances the model's ability to generalize across the various categories without introducing bias toward any specific class.

After completing the dataset separation, data augmentation techniques are applied to the training set. The augmentation method includes adding distortions, rotations, and horizontal and vertical flips. The procedure of resizing and data augmentation is shown in the figure below.

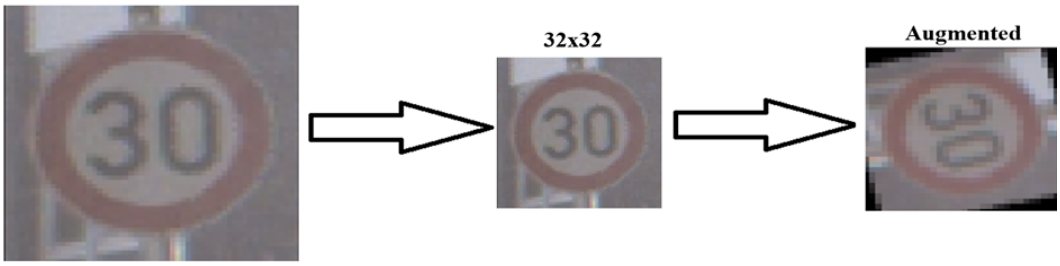


Figure 2: Resized and augmented training sample

The test samples are manually labeled using the provided .csv file from the dataset zip. To maintain a 70-15-15 split for training, validation, and testing, 1,290 test images—the same quantity as in the validation dataset—are randomly selected from the raw Test.zip to create the final test set.

The unseen new dataset for demonstration is also randomly selected from the raw Test.zip. some new distortions are added to the sampled new dataset to make sure it is never seen by the model. All the images are transformed to size 32 by 32, to make sure the input dataset size is consistent, and the images are easy to read. Moreover, the input images are normalized to guarantee the model learns more easily and gives better results.

4 ARCHITECTURE

As shown in the illustration figure, our current model consists of an initial Color Filter Layer followed by a modified GoogLeNet structure. The Color Filter Layer has 64 filters, which is a standard choice for deep CNNs as it provides balanced feature extraction with efficient computational cost. This layer captures key color and texture features early on, which are essential in traffic sign classification. Each filter learns specific patterns or colors from the RGB input, contributing a diverse set of initial features that enhance the model's ability to differentiate traffic signs in complex environments. Before entering the Inception modules, the network includes initial convolutional and max-pooling layers. These downsample the input images and extract basic features like edges and textures, making the network's processing of intermediate features in later stages more effective.

The model employs two Inception blocks: Inception 1A and 2A.

- **Inception 1A**: Operates on inputs with larger spatial dimensions and fewer channels, extracting lower-complexity intermediate features.
- **Inception 2A**: Processes inputs with reduced spatial dimensions but more feature channels. This block captures high-level, abstract features such as symbols, shapes, and intricate patterns specific to traffic signs.

To prevent overfitting, we first decrease inception blocks from 4 to 2. In each block, we integrated by applying batch normalization after the activation function, which is similar to inception V2. After each Inception module, a max-pooling layer further reduces spatial dimensions and highlights critical features. An adaptive average pooling layer then reduces the output to 1x1 spatial dimensions, making it ready for the final fully connected layer, which performs the classification. Our model is trained with learning rate equals to 0.001, 32 batch size and 30 epoches, which allowed it to learn effectively without overfitting.

5 BASELINE MODEL: LENET-5

We will use LeNet-5 as our baseline model, which is a classic convolutional neural network (CNN) architecture that consists of 2 convolutional layers, 2 pooling layers, and 2 fully connected layers. It

was an early CNN that is relatively simple and originally designed for grayscale images (MNIST), so its architecture doesn't take advantage of color features, which are essential in traffic sign classification. It extracts low-level features from the input image, downsamples them through pooling, and then classifies the features into one of the categories. While LeNet-5 is computationally efficient and easy to replicate, its shallow architecture limits its ability to handle complex real-world tasks such as traffic sign recognition (TSR), where modern models can capture multi-scale features and efficiently process color information.

LeNet-5 is a small network, which can classify traffic signs, but it does not handle the complexity of real-world traffic sign datasets well for the following reasons:

- **Small and Shallow Network**: LeNet-5 only has 2 convolutional layers followed by pooling layers, making it shallow in comparison to more modern architectures like Inception. It may struggle with the diverse scales and variations of traffic signs in real environments.
- **Limited Feature Extraction**: The small kernel sizes and limited layers restrict the model's ability to extract fine-grained features such as subtle shape variations and multi-scale features.
- **Poor Handling of Color Information**: LeNet-5 was originally designed for grayscale images (MNIST), so its architecture doesn't take advantage of color features, which are essential in traffic sign classification.
- **Low Real-time Performance**: Due to its architecture, LeNet-5 is inefficient in real-time recognition tasks, especially for large-scale datasets.

By using the Inception Network (GoogLeNet), our project aims to outperform LeNet-5 in several key areas:

- **Multi-Scale Feature Extraction**: Inception modules allow our model to capture features at different scales (small, medium, large), making it better suited to handle the variety of shapes and sizes in traffic signs.
- **Efficient Model Size**: Despite being deeper, the Inception Network reduces the number of parameters using clever architectural design (1x1 convolutions). This keeps the model lightweight while increasing its ability to classify complex images efficiently.
- **Color Feature Handling**: Our project extracts and processes the color information explicitly, which is crucial for distinguishing between traffic signs that may share similar shapes but have different colors.
- **Real-Time Capability**: By focusing on reducing the model size and increasing efficiency, our project could enable real-time traffic sign recognition on vehicles, which is essential for autonomous driving applications.

6 Quantitative Results

The performance of our traffic sign recognition model was tested on a suite of quantitative metrics so as to determine comprehensively its accuracy, generalization ability, and classification behavior. The major results are given below.

6.1 Training, Validation, and Test Accuracy

The model had a very consistent rise in training with the reduction of training and validation loss almost constantly over 30 epochs (Fig 3). This signifies effective learning with no conspicuous overfitting. The final test accuracy on the GTSRB dataset was 93.64%, showing strong generalization to unseen data.

6.2 Precision, Recall, and F1-Score

These metrics further detail the class-wise performance of the model (Fig4):



Figure 3: Training and Validation Loss and Accuracy over 40 Epochs for our final model

	precision	recall	f1-score	support
0	0.83	1.00	0.91	5
1	0.96	0.91	0.93	77
2	1.00	0.74	0.85	74
3	1.00	0.86	0.93	37
4	0.91	0.94	0.92	64
5	0.70	0.97	0.82	71
6	1.00	1.00	1.00	10
7	0.80	0.91	0.85	35
8	0.98	0.84	0.90	49
9	1.00	1.00	1.00	44
10	1.00	1.00	1.00	64
11	0.98	1.00	0.99	44
12	1.00	0.99	0.99	72
13	0.99	1.00	0.99	89
14	1.00	1.00	1.00	18
15	0.88	1.00	0.94	22
16	1.00	1.00	1.00	13
17	1.00	0.90	0.95	31
18	1.00	0.79	0.88	43
19	1.00	1.00	1.00	6
20	0.88	1.00	0.94	15
21	0.50	0.44	0.47	9
22	0.60	1.00	0.75	6
23	1.00	0.75	0.86	20
24	1.00	1.00	1.00	10
25	0.99	0.97	0.98	68
26	0.91	1.00	0.95	20
27	1.00	1.00	1.00	5
28	1.00	0.91	0.95	11
29	0.92	1.00	0.96	11
30	0.89	1.00	0.94	16
31	0.94	1.00	0.97	32
32	1.00	1.00	1.00	6
33	1.00	0.91	0.95	22
34	0.86	1.00	0.92	18
35	0.89	0.94	0.91	34
36	1.00	0.94	0.97	17
37	0.75	1.00	0.86	6
38	0.99	0.97	0.98	68
39	0.86	1.00	0.92	6
40	1.00	0.91	0.95	11
41	1.00	1.00	1.00	5
42	1.00	1.00	1.00	6
accuracy			0.94	1290
macro avg	0.93	0.94	0.93	1290
weighted avg	0.95	0.94	0.94	1290

Figure 4: Class-wise Precision, Recall, F1-Score

- **Macro Average:** Precision = 0.93, Recall = 0.94, F1-Score = 0.93. It reflects the well-balanced performance across all classes.
- **Weighted Average:** Precision = 0.95, Recall = 0.94, F1-Score = 0.94. This accounts for class imbalance and thus presents the robustness of the model for most classes.

Most classes have a very high F1-score, proving to be very good performance, but the lower F1-scores in certain underrepresented classes, such as Class 21, hint at further scope of improvement.

6.3 Confusion Matrix

The confusion matrix (Fig5) provides insights into class-wise errors. The misclassifications usually happened between the most visually similar signs (e.g., Speed Limit signs : Class 0-8), and therefore indicating the need for improved fine-grained feature extraction. This matrix reveals the overall good performance with diagonal dominance, which indicates the model's overall excellent performance. Off-diagonal elements are indicative of instances where poor quality or ambiguity introduced challenges in classification.

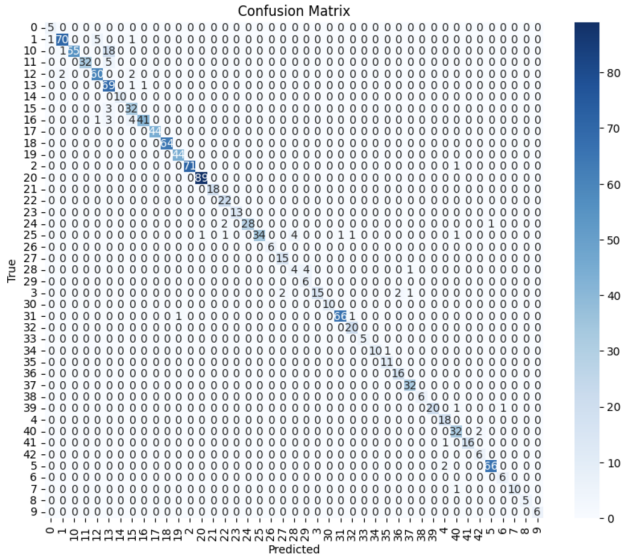


Figure 5: Confusion Matrix showing class-wise predictions and misclassifications for the model

7 Qualitative Results

In addition to the quantitative results, qualitative analysis complemented this quantitative evaluation. Through sample outputs, it looked at the strengths and pointed toward limitations.

7.1 Correct Classifications

The model performs well on clear, high-resolution signs with distinct shapes and colors (Fig6), demonstrating its ability to extract meaningful features and leverage color information effectively.

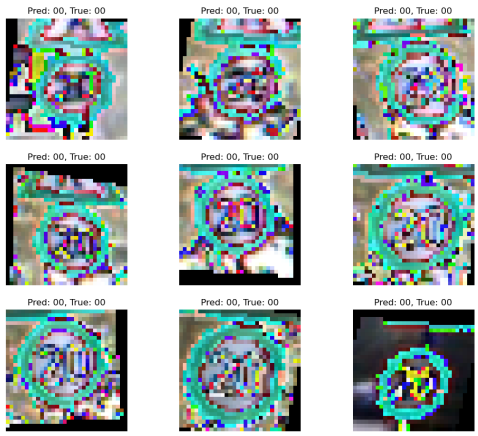


Figure 6: Successfully classified traffic sign images with true and predicted labels with our final model

7.2 Misclassified Images

Other mistakes that were seen include the following (Fig7):

- **Visually Similar Signs**: Confusions such as between ”Speed Limit 30 km/h” and ”Speed Limit 50 km/h” were common.
- **Low-Quality Images**: Poor lighting, motion blur, or partial occlusion affected the model's ability to extract meaningful features.

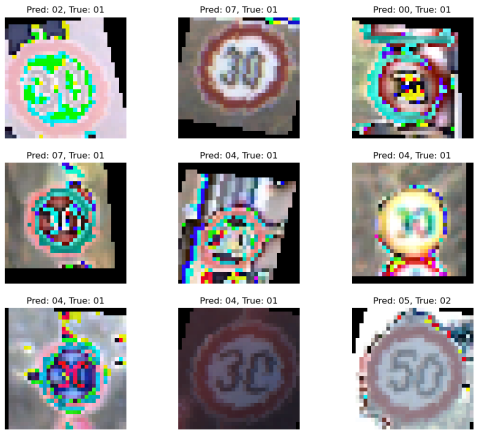


Figure 7: Misclassified traffic sign images with true and predicted labels with our final model

Therefore, such errors thus mark the points of refinement, such as data augmentation or the adoption of advanced architectures to be able to handle complex multi-scale features.

7.3 Insights and Context

The qualitative results align with the quantitative findings. The confusion matrix and F1-Score reflect the model's difficulty with specific classes, which is corroborated by the misclassified examples. Some examples being that speed limit signs get misclassified since there is a very small difference in text or in the design of the sign that is not properly identified because a better texture and fine-grained feature set recognition was needed.

8 Evaluate the model on new data

To test the performance our model on new data, our team used The German Traffic Sign Detection Benchmark (GTSDB) as our test dataset. Houben et al. (2013) The dataset consists of 900 images of German Traffic Sign with the same sign type to GTSRB, and thus is an ideal test subject. To obtain a more rigorous and realistic test result, we did affine transformation to the datasets to simulate incomplete or cut and rotated sign images that may be obtained in real driving environments.

As a result, our model gets the accuracy of 77.49% on the test new data, which is significantly lower than the average level during training and testing. Without the interference of the data transformation, the model's final accuracy reaches 88.29%, which is 10% below the average level.

8.1 Correct Classifications

The model performs well on distinguish the the outer contour and main color blocks (Fig8), especially for non-numeric signs.

8.2 Misclassified Images

Other mistakes on test new data that were seen include the following (Fig9):

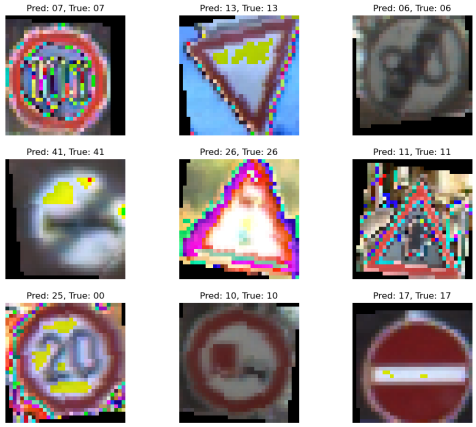


Figure 8: Successfully classified test traffic sign images

- **Numeric Signs**: Confusions between different numbers are common.
- **Inverted Similar Sign**: Confusions between inverted similar signs, such as ”turn left” and ”turn right” are common.



Figure 9: Misclassified test traffic sign images

9 Discussion on result

Generally, our model's performance reached our teams expectation on test data. Comparing to 95% weighted average evaluate accuracy, the 77% test accuracy strongly proved the ability of our model to identify and extract embeddings from transformed unfamiliar data. The generalization ability of our model has been shown to be excellent: the model could successfully identify different shape and large blocks of color with nearly 100% accuracy. However, our model's performance on some details and recognition difficulties is unsatisfactory.

During train, the model did not do well on distinguishing numeric signs. Thus, it is predictable to the relatively low accuracy on numbers for test new data. Our team supposed its main reason to be the low-resolution image to the input layer. Different from the ideal handwritten digit recognition, the uneven lighting conditions in reality, as well as the image information limited by the circle, lead to that the 32x32 input layer is insufficient to remain enough traits for the model to identify numbers. We believe that the model would perform much better with a higher input resolution.

Another interesting fact is that our model could not reach high accuracy on signs that are flipped and symmetrical to each other. Using the example of various signs that allow turns and straights, the tail of the white arrow on the blue background may not be correctly identified and only the arrowhead's trunk was successfully identified. This is also considered due to the uneven lighting and shooting conditions.

10 Ethical Considerations

10.1 Privacy and Copyright

The Dataset used for the project conforms CC0 1.0 Universal code, which means it is in the public domain and has no copyright. The photographs in the dataset only include traffic signs and has no private information.

10.2 Regional bias and limitation

The newest dataset used for the project is from German and was published in 2013. It means that the result for the project is only applicable to common traffic signs in German traffic conditions, and is not responsible for any sign changes since 2013.

10.3 Responsibility and Safety

To avoid underlying responsibility and safety issues, our team guarantee that the project result is only for study and research purpose. It is stated that the project result model cannot be and will not be used in real traffic conditions.

11 Project Difficulty / Quality

The traffic sign recognition project presents notable challenges due to the complex nature of real-world environments, such as camera resolution, lighting, and weather, so the traffic signs can vary significantly in size, color, and orientation. These factors make it difficult to accurately identify and classify signs under diverse scenarios.

Instead of leveraging CNNs to solve the problem, an improved version of GoogLeNet is implemented. The team added an extra color filter to the model. As mentioned before, the hue and brightness of input samples varies a lot, so the well-designed color filter improves the model's sensitivity to the distinctive hues of traffic signs. Also, the inception layers are designed with extra batch normalization layers ,making it Inception V2, to extract both low-complexity, intermediate features, and high-level features, that are crucial for distinguishing symbols, shapes, and patterns. The hyperparameters of these layers are carefully tuned to ensure the best performance.

The team encountered many challenges during the training process. Due to the fact that the training process requires significant GPU capacity, the team ran it locally. The validation and test results implied that the model was overfitting in the beginning. Many misclassified samples showed blurred or distorted shapes due to lighting issues. To solve this problem, new augmentation methods on the hue, lighting, and brightness of the training dataset. With this improvement, the model performs a great job with a test accuracy of 95%, and the number is comparable to the state-of-the-art results.

12 Link to Github Repo

https://github.com/jassiyu/APS360-Group6-Project

REFERENCES

X. W. Gao, L. Podladchikova, D. Shaposhnikov, K. Hong, and N. Shevtsova. Recognition of traffic signs based on their colour and shape features extracted using human vision models, 2006. https://doi.org/10.1016/j.jvcir.2005.10.003.

Sebastian Houben, Johannes Stallkamp, Jan Salmen, Marc Schlipsing, and Christian Igel. Detection of traffic signs in real-world images: The German Traffic Sign Detection Benchmark. In *International Joint Conference on Neural Networks*, number 1288, 2013.

Chang-il Kim, Jinuk Park, Yongju Park, Woojin Jung, and Yong-seok Lim. Deep learning-based real-time traffic sign recognition system for urban environments. *Infrastructures*, 8(2):20, 2023. https://doi.org/10.1016/j.prime.2024.100442.

Yan Li, Zhiqiang Xie, Xiaodong Deng, and Yuting Wu. Traffic sign detection based on improved faster r-cnn for autonomous driving. *Journal of Supercomputing*, 2023. https://doi.org/10.1007/s11227-021-04230-4.

X. Qiao. Research on traffic sign recognition based on cnn deep learning network. *Procedia Computer Science*, 228:826–837, 2023. https://doi.org/10.1016/j.procs.2023.11.102.

J. Shen, Z. Zhang, J. Luo, and X. Zhang. Yolov5-ts: Detecting traffic signs in real-time. *Frontiers in Physics*, 11, 2023. https://doi.org/10.3389/fphy.2023.1297828.

Ming Tang. A real-time traffic sign detection in intelligent transportation system using yolov8-based deep learning approach. *Signal, Image and Video Processing*, 18:6103–6113, 2024. https://doi.org/10.1007/s11760-024-03300-3.

Yihui Wu, Yulong Liu, Jianmin Li, Huaping Liu, and Xiaolin Hu. Traffic sign detection based on convolutional neural networks. In *The 2013 International Joint Conference on Neural Networks (IJCNN)*, pp. 1–7. IEEE, 2013. https://doi.org/10.1109/IJCNN.2013.6706811.

N. Youssouf. Traffic sign classification using cnn and detection using faster-rcnn and yolov4. *Heliyon*, 8(12):e11792–e11792, 2022. https://doi.org/10.1016/j.heliyon.2022.e11792.